Vivekanand Education Society's Institute of Technology, Chembur, Mumbai,
Department Of Computer
Academic Year 2025-26
MIDTERM TEST

Class : D12	**Division: A / B/ C**
Semester: VI	**Subject: Cryptocurrency and Blockchain Development**
Date: 9th March 2026	**Time: 9:00 am to 10:00 am**

Q1.a. Discuss the benefits of immutability properties in Blockchain

Immutability means that once data is written to a blockchain, it **cannot be altered or deleted**.

1. **Data Integrity :** Ensures that the stored data remains accurate and unchanged.
2. **Trust and Transparency :** All participants can verify transactions without relying on a central authority.
3. **Fraud Prevention :** Prevents tampering with transaction records.
4. **Auditability :** Provides a permanent and verifiable transaction history.
5. **Security :** Cryptographic hashing links blocks together, making unauthorized modifications extremely difficult.
6. **Decentralized Verification :** Every node maintains a copy of the ledger, reducing single points of failure.

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Q1.b. Gopal has 5 ETH in his wallet. He transfers 1.1 ETH to Ram and pays gas of 0.002 ETH. And later spends 0.6 ETH to Geeta's account by paying 0.007 ETH. Find Gopal's wallet balance.

Initial balance (Gopal) = **5 ETH**

Transaction 1 : Transfer to Ram = **1.1 ETH**
Gas fee = **0.002 ETH**
Total spent = 1.1 + 0.002 = 1.102 ETH
Balance = 5 − 1.102 = 3.898 ETH

Transaction 2 : Transfer to Geeta = **0.6 ETH**
Gas fee = **0.007 ETH**
Total spent = 0.6 + 0.007 = 0.607 ETH
Balance = 3.898 − 0.607 = 3.291 ETH

Final Balance(Gopal) = 3.291 ETH

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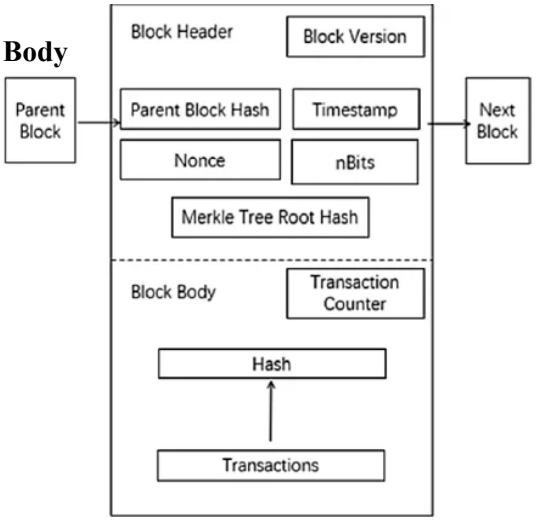
Q1. c. With the help of a neat diagram, demonstrate the structure of Block in Blockchain

A block contains two main parts: (1) **Block Header and (2) Block Body**
Previous Block Hash – links blocks together

- **Merkle Root** – hash of all transactions
- **Nonce** – used in mining
- **Timestamp** – block creation time
- **Difficulty Target** – mining difficulty level



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Q1.d. Identify the key functions to be written for ERC20 token

An **ERC20 token** is a standard for creating tokens on the Ethereum blockchain.

1. **totalSupply() :** Returns total number of tokens.
2. **balanceOf(address account) :** Returns token balance of a specific account.
3. **transfer(address recipient, uint amount) :** Transfers tokens to another address.
4. **approve(address spender, uint amount) :** Allows another address to spend tokens.
5. **allowance(address owner, address spender) :** Returns remaining tokens allowed to spend.
6. **transferFrom(address sender, address recipient, uint amount) :** Transfers tokens on behalf of another account.

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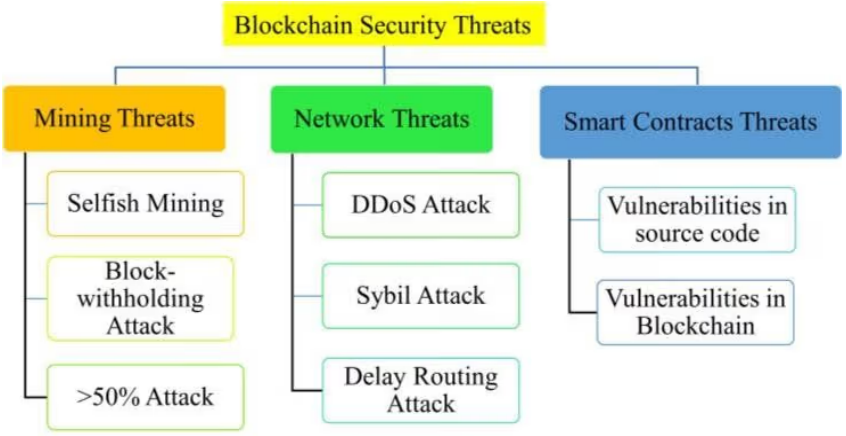
Q1e. Enlist the security threats in the Blockchain Ecosystem

- **51% Attack :** A miner controls more than half of the network hashing power.
- **Double Spending Attack :** Same cryptocurrency is spent more than once.
- **Sybil Attack :** Creation of multiple fake nodes to influence the network.
- **Smart Contract Vulnerabilities :** Bugs in contract code exploited by attackers.
- **Phishing Attacks :** Fake websites steal private keys.
- **Private Key Theft :** Unauthorized access to wallets.
- **Routing Attacks :** Intercepting blockchain network traffic.
- **Eclipse Attack :** Isolating a node and feeding it false data.



Q1.f. Given a simplified hash function as given below hash = (txid + nonce) % 1000
Find the smallest nonce for the transaction id: 683 that satisfies the mining rule: hash < 500.

<u>Given</u> : txid = 683
Mining rule: hash < 500

<u>Note</u> : The smallest value occurs when the result becomes 0–499.
So, the next wrap around occurs at 1000
That is, 683 + nonce = 1000
Nonce = 317
Now, Hash = (683+317) % 1000 = 1000 % 1000 = 0
Therefore, the **Smallest nonce = 317**

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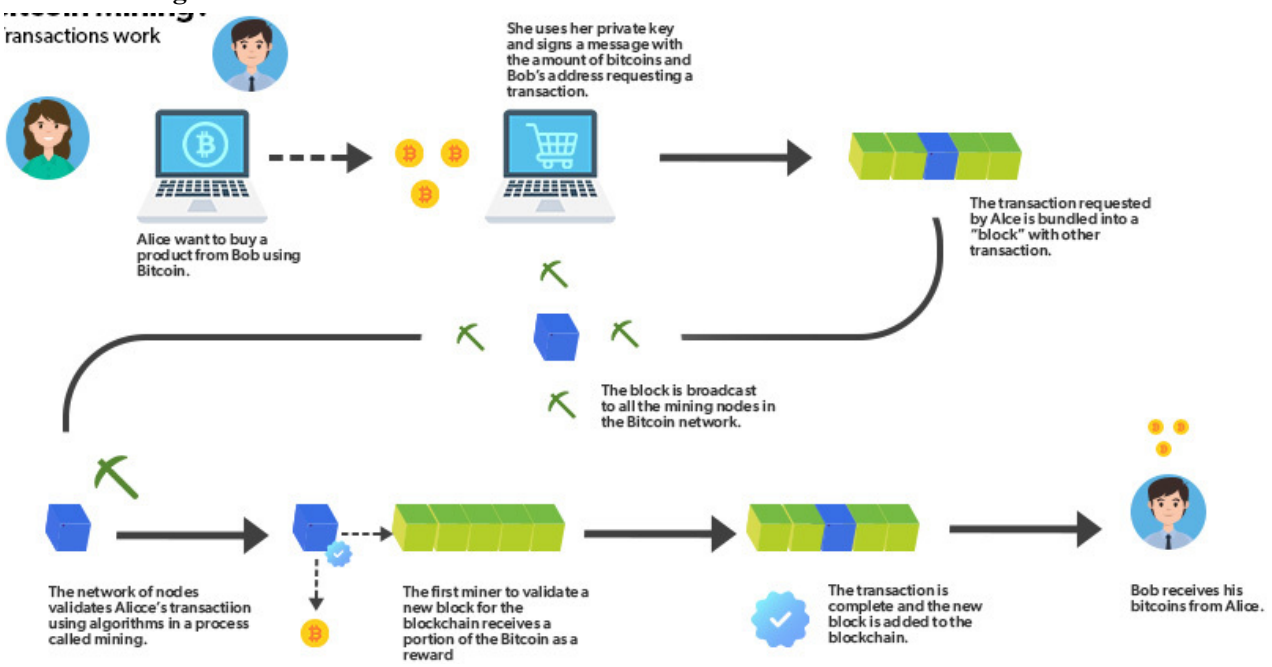
Q2. a. Discuss the process of mining based on the Proof of Work consensus mechanism in Bitcoin

Mining Steps

1. **Transaction Broadcast :** Users broadcast transactions to the network.
2. **Transaction Verification :** Nodes verify signatures and balances.
3. **Transaction Pool :** Verified transactions enter the mempool.
4. **Block Formation :** Miners collect transactions into a candidate block.
5. **Hash Puzzle Solving :** Miners repeatedly change the **nonce** to produce a valid hash.
6. **Difficulty Requirement :** Hash must be lower than the target value.
7. **Block Broadcast :** Winning miner broadcasts the block to the network.
8. **Block Validation :** Other nodes verify and add it to the chain.
9. **Mining Reward :** Miner receives block reward + transaction fees.



Q2. b. Explain the possible threats to crypto wallets. Also, discuss the best practices to safeguard the Crypto wallets

Threats

1. Phishing attacks
2. Malware / Keyloggers
3. Private key leakage
4. Exchange hacks
5. SIM swapping
6. Fake wallet apps
7. Social engineering attacks

Best Practices

1. Use Hardware Wallets
2. Enable Two-Factor Authentication (2FA)
3. Keep Private Keys Offline
4. Backup Seed Phrase Securely
5. Use Strong Passwords
6. Avoid Public Wi-Fi for Transactions
7. Download wallets only from official sources
8. Regularly update wallet software

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Q3. a. Compare the Blockchain Platforms Bitcoin and Ethereum in a tabular format with parameters

Parameter	Bitcoin	Ethereum
Launch Year	2009	2015
Founder	Satoshi Nakamoto	Vitalik Buterin
Purpose	Digital currency	Smart contract platform
Consensus	Proof of Work	Proof of Stake
Programming Capability	Limited scripting	Turing complete smart contracts
Block Time	~10 minutes	~12 seconds
Native Currency	BTC	ETH
Smart Contracts	Not supported	Fully supported
Applications	Digital payments	DeFi, NFTs, DApps

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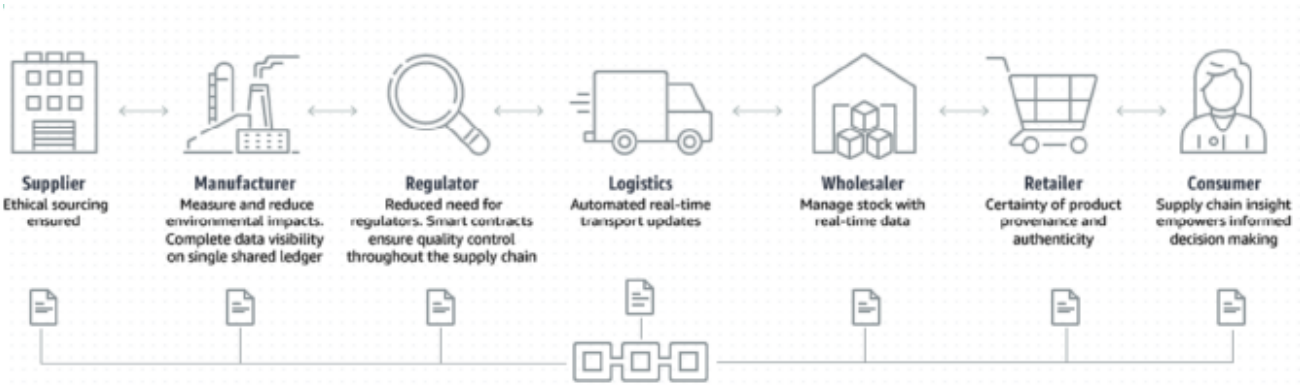
Q3. b. With a neat diagram, explain how Blockchain can be integrated into Supply Chain Management to ensure integrity, transparency and traceability.

Blockchain provides **transparent and tamper-proof tracking of goods** across the supply chain.

Working

1. Supplier records raw material origin.
2. Manufacturer records production details.
3. Distributor logs shipping information.
4. Retailer records inventory arrival.
5. Customer verifies product authenticity.

Benefits

- Transparency
- Traceability
- Reduced fraud
- Improved accountability
- Real-time tracking